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ARTICLE

Algorithm 543: FFT9, Fast Solution of Helmholtz-Type Partial Differential Equations [D3]

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ALGORITHM 543

FFT9, Fast Solution of Helmholtz-Type Partial Differential Equations [D3]

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CR Categories: 5 17

Language: Fortran

DESCRIPTION

The algorithm given here is a complement to [1] where the description, test results, and references are given.

REFERENCES

1. HOUSTIS, E.N., AND PAPATHEODOROU, T.S. High-order fast elliptic equation solver. *ACM Trans. Math. Software* 5, 4 (Dec. 1979), 431-441.

ALGORITHM

[Summary information and part of the listing is printed here. The complete listing is available from the ACM Algorithms Distribution Service (see inside back cover for order form), or may be found in "Collected Algorithms from ACM."]

NAME(n): indicates a Fortran module with n records

NAME^T(n): indicates "NAME" is included for testing purposes

NAME^D(n): indicates "NAME" contains test data

Contents: MAIN(299), RGHTSD(208), EQSOL(77), DISCRT(95), EV-ENRD(55), ODDRD(54), CRED(76), STOREX(18), FETCHX(18),

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KFOLD(138), FOUR(51), NEG(12), ZERO(13), TFOLD(16),
 SETF(41), SUMARY(76), PDERGH^T(7), BCOND^T(7),
 TRUE^T(5), PDE^T(9), DATA^D(3)

```

C      PROGRAM FFT9(INPUT,OUTPUT,TAPE5=INPUT,TAPE6=OUTPUT)      10
C                                                                20
C      ---- PROGRAM DESCRIPTION ----                             30
C      PROGRAM FFT9 USES A 4-TH OR 6-TH ORDER 9-POINT DIFFERENCE 40
C      FORMULA AND FAST FOURIER TRANSFORM FOR THE NUMERICAL      50
C      SOLUTION OF THE ELLIPTIC EQUATION WITH CONSTANT COEFFICIENTS 60
C                                                                70
C      (I) CUXX*DDXU + CUY*DDYU + CU*U = R                       80
C                                                                90
C      ON A RECTANGULAR REGION 0 .LE. X .LE. SX, 0 .LE. Y .LE. SY 100
C      AND SUBJECT TO DIRICHLET BOUNDARY CONDITIONS              110
C      (II) U = G                      ON THE BOUNDARY            120
C                                                                130
C      NOTE- THE 6-TH ORDER ALGORITHM IS APPLIED ONLY           140
C      TO POISSON TYPE OPERATORS                                 150
C                                                                160
C                                                                170
C      ---- INPUT AND OUTPUT TO FFT9 ----                        180
C      --PROBLEM DEFINITION-- USER SUPPLIED FORTRAN FUNCTION FOR THE 190
C      EVALUATION OF THE RIGHT SIDES (R,G) OF THE DIFFERENTIAL    200
C      AND BOUNDARY OPERATORS                                    210
C                                                                220
C      REAL          FUNCTION PDERGH(X,Y)                        230
C                                                                240
C      PDERGH = R                                               250
C      RETURN                                                    260
C      END                                                        270
C      REAL          FUNCTION BCOND(I,X,Y,BVALUS)                280
C      TWO DIMENSIONS                                           290
C      VALUES OF BOUNDARY CONDITION ON SIDE I                  300
C      AT (X,Y)                                                  310
C                                                                320
C      I=4                                                        330
C      -----                                                  340
C      I          I                                              350
C      I          I                                              360
C      I=3 I      REGION      I      I=1                        370
C      I          I                                              380
C      I          I                                              390
C      -----                                                  400
C      I=2                                                        410
C                                                                420
C      REAL          BVALUS(4)                                    430
C      GO TO(100,101,102,103) , I                                440
C      101 BVALUS(4) = G                                          450
C      BCOND = BVALUS(4)                                         460
C      RETURN                                                    470
C      102 BVALUS(4) = G                                          480
C      BCOND = BVALUS(4)                                         490
C      RETURN                                                    500
C      103 BVALUS(4) = G                                          510
C      BCOND = BVALUS(4)                                         520
C      RETURN                                                    530
C      104 BVALUS(4) = G                                          540
C      BCOND = BVALUS(4)                                         550
C      RETURN                                                    560

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C          END 570
C 580
C  USER SUPPLIED SUBROUTINE FOR THE DEFINITION OF 590
C  P.D.E CONSTANT COEFFICIENTS 600
C 610
C  SUBROUTINE PDE(X,Y,CVALUS) 620
C 630
C  REAL CVALUS(7) 640
C  CVALUS(1) = CUXX 650
C  CVALUS(3) = CUYX 660
C  CVALUS(6) = CU 670
C  RETURN 680
C  END 690
C 700
C 710
C  USER SUPPLIED FORTRAN FUNCTION FOR THE TRUE SOLUTION 720
C  IF KNOWN 730
C 740
C  REAL FUNCTION TRUE(X,Y) 750
C  TRUE = ... 760
C  RETURN 770
C  END 780
C 790
C  --REGION AND GRID SPECIFICATIONS-- 800
C 810
C  SX,SY - LENGTHS OF SIDES OF RECTANGLE 820
C  NGRIDX,NGRIDY - NUMBER OF HORIZONTAL AND 830
C  VERTICAL MESH LINES 840
C  IQX,IQY - EXPONENTS OF 2 850
C  NGRIDX=2**IQX+1,NGRIDY=2**IQY+1 860
C 870
C  READ(5,100) SX,SY,NGRIDX,NGRIDY 880
C  100 FORMAT(2F10.0,2I3) 890
C 900
C  --OUTPUT CONTROL--USER SUPPLIED DATA 910
C  LEVEL - OUTPUT LEVEL DESIRED 920
C  NRUNS - NUMBER OF SUCCESSIVE RUNS. 930
C  IN EACH RUN THE MESH SIZE IS 940
C  CUT BY A FACTOR OF 2. 950
C  ORDER - RATE OF CONVERGENCE DESIRED 960
C  READ(5,102) LEVEL,NRUNS,ORDER 970
C  102 FORMAT(3I2) 980
C 990
C  IF LEVEL = 0 PRINT APPROXIMATE SOLUTION AT NODES 1000
C  = 1 PRINT MAXIMUM ERROR AND MAXIMUM 1010
C  RELATIVE ERROR PROVIDED TRUE SOLUTION 1020
C  IS KNOWN 1030
C  = 2 ALSO PRINT TRUE SOLUTION AND 1040
C  APPROXIMATE SOLUTION AT THE INTERIOR 1050
C  GRID POINTS 1060
C 1070
C  IF ORDER = 4 CHOOSE A 4-TH ORDER DIFFERENCE APPR. TO (I) 1080
C  = 6 CHOOSE A 6-TH ORDER DIFFERENCE APPR. TO (I) 1090
C  --STORAGE--IT IS ASSUMED THAT 3 .LE. IXQ,IQY .LE. 7 . 1100
C  IN CASE OF FINER MESH THE DIMENSIONS REQUIRED ARE 1110
C  COMPUTED BY FORMULAS GIVEN IN ARRAY LIST DESCRIPTION. 1120
C  ---- MAIN VARIABLES OF FFT9 ---- 1130
C 1140
C  WORK - WORKING SPACE OF DIMENSION 1150
C  ORDER = 6 7+10*MAX(NGRIDX,NGRIDY)+2*NGRIDX*NGRIDY 1160
C  Z,Y - ARRAYS USED IN THE FOURIER ANALYSIS-SYNTHESIS 1170
C  ORDER = 4 7+10*MAX(NGRIDX,NGRIDY)+NGRIDX*NGRIDY 1180

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C		DIMENSION OF 'Z,Y' IS NZD=NYD= MAX(NX,NY)	1190
C	AKX(I+1)	-THE I-TH EIGENVALUE OF THE DIAGONAL BLOCK DIVIDED BY	1200
C		THE I-TH EIGENVALUE OF THE OFF DIAGONAL BLOCK	1210
C		SQUARED MINUS 2	1220
C		DIMENSION OF 'AKX' IS NAKXD = MAX(NX,NY)+2	1230
C			1240
C			1250
C	CORE(K)	- VALUE OF APPROXIMATE SOLUTION AT NODE K	1260
C		DIMENSION OF 'CORE' IS NCORED = NX+2+(NX+1)*(NY+1)	1270
C	PTINT()	- VALUES OF G AT HALF LATTICE POINTS	1280
C		DIMENSION IS NPINTD = NX*NY	1290
C	NOTE-	ARRAY 'PTINT' IS NOT USED IN CASE ORDER .EQ. 4	1300
C			1310
C	GRIDX,GRIDY	- GRID COORDINATES	1320
C		DIMENSION OF 'GRIDX,GRIDY' IS	1330
C		NGRDXD = NX+1,NGRDYD = NY+1	1340
C			1350
C	SX,SY	- LENGTHS OF RECTANGULAR REGION	1360
C			1370
C	ED(I+1)	- THE I-TH EIGENVALUE OF THE OFF DIAGONAL	1380
C		BLOCK.DIMENSION OF 'ED' IS NEDD = MAX(NX,NY)+2	1390
C	INDEX	- INDEX VECTOR USED IN FFT ANALYSIS	1400
C		DIMENSION IS INDEXD = MAX(NX,NY)	1410
C	SI	- SINES VECTOR FOR FFT ANALYSIS AND	1420
C		SYNTHESIS. DIMENSION NSID = MAX(NX,NY)	1430
C			1440
C	----	COMMON VARIABLES OF FFT9 ----	1450
C			1460
C	COMMON /MESH/		1470
C	NX,NY	- NX = 2**IQX,NY = 2**IQY	1480
C	IMIN,IMAX	- RANGE OF INTERIOR NODES IN X-DIRECTION	1490
C	JMIN,JMAX	- RANGE OF INTERIOR NODES IN Y-DIRECTION	1500
C	INC	- INC = IMAX - IMIN + 3	1510
C	IRO	- IRO = NX + 3	1520
C	IBCX	- IBCX = 1	1530
C	IQX	- EXPONENT OF 2	1540
C	IBCY	- IBCY = 1	1550
C	IQY	- EXPONENT OF 2	1560
C	HX,HY	- MESH SIZE	1570
C	HXY2	- HXY2 = (HX/HY)**2	1580
C	PI	- PI = 3.14...	1590
C	POTFAC	- NORMALIZATION FACTOR = 2/NX	1600
C	COMMON /FDFORM/		1610
C	DLEFT	- DIAGONAL ENTRY OF THE OFFDIAGONAL BLOCK	1620
C	DIAG,OFFD	- DIAGONAL AND OFFDIAGONAL ENTRIES OF THE	1630
C		DIAGONAL BLOCK OF THE 9-POINT FORMULA	1640
C	RF,HH,CH,RC1,FACTOR	- CONSTANTS	1650